

Georgia Institute of Technology
School of Aerospace Engineering

AE 6114 — Fundamentals of Solid Mechanics — Fall 2022

Quiz No.1

Problem 1. (40 Points)

Consider a cube of material of length L , the body occupies the space $\mathcal{B} = \{(X_1, X_2, X_3) : 0 \leq X_1 \leq L, 0 \leq X_2 \leq L, 0 \leq X_3 \leq L\}$ in the reference configuration.

The body undergoes a motion defined by

$$\chi(\mathbf{X}) = \left(\frac{a}{L}X_1 + \gamma X_2\right)\mathbf{e}_1 + X_2\mathbf{e}_2 + X_3\mathbf{e}_3$$

where a and γ are given constants.

- (a) Compute the deformation gradient $\mathbf{F}$, the Right Cauchy-Green tensor $\mathbf{C}$ and the Green strain tensor $\mathbf{E}$.
- (b) Compute the displacement field $\mathbf{u}$ and the infinitesimal strain tensor $\boldsymbol{\epsilon}$.
- (c) Using **both** finite and infinitesimal deformations, calculate the stretch λ in for fibers aligned with the $\mathbf{e}_1$ -direction.
- (d) Using **both** finite and infinitesimal deformations, calculate the change in angle θ between two fibers oriented in the $\mathbf{e}_1$ and $\mathbf{e}_2$ directions.
- (e) Under what conditions is the infinitesimal strain $\boldsymbol{\epsilon}$ an appropriate measure of strain for this motion? For these conditions, show that $\mathbf{E}$ reduces to $\boldsymbol{\epsilon}$ and that your stretch and angle measurements also coincide.

Problem 2. (40 Points)

For the same body described above, consider the motion

$$\chi(\mathbf{X}) = (aX_1 - bX_2)\mathbf{e}_1 + (bX_1 + aX_2)\mathbf{e}_2 + X_3\mathbf{e}_3$$

where a and b are **constants**.

- (a) Compute the Right Cauchy-Green tensor $\mathbf{C}$, the stretch tensor $\mathbf{U}$, and the rotation tensor $\mathbf{R}$. Recall that $\mathbf{F} = \mathbf{R}\mathbf{U}$ and hence $\mathbf{R} = \mathbf{F}\mathbf{U}^{-1}$.
- (b)
 - (i) What relationship between the constants a and b must be satisfied in order for $\mathbf{F} = \mathbf{R}$?
 - (ii) If this relationship is satisfied (i.e. $\mathbf{F} = \mathbf{R}$), what kind of deformation is described by the motion χ given?
- (c) Assume that a and b are constrained such that $\mathbf{F} = \mathbf{R}$. **Compute** the infinitesimal strain tensor $\boldsymbol{\epsilon}$ and **comment** on whether the result makes physical sense.

Problem 3. (20 Points)

- (a) Given $\underline{\underline{\mathbf{A}}}$ a **skew** tensor, and $\underline{\mathbf{b}}$ an arbitrary vector, show that $\underline{\mathbf{b}} \cdot \underline{\underline{\mathbf{A}}} \cdot \underline{\mathbf{b}} = 0$.
- (b) Given $\underline{\underline{\mathbf{A}}} = (\underline{\mathbf{a}} \times \underline{\mathbf{b}}) \otimes \underline{\mathbf{c}}$ and $\underline{\mathbf{a}} \otimes \underline{\mathbf{c}} = \underline{\underline{\mathbf{1}}}$, show that $\underline{\underline{\mathbf{A}}}$ is **skew**.
- (c) Show that the stretch in a given direction is frame indifferent. That is, the operation $\underline{\mathbf{N}} \cdot \underline{\underline{\mathbf{C}}} \cdot \underline{\mathbf{N}}$ gives the same result independently of the frame adopted to represent the tensor $\underline{\underline{\mathbf{C}}}$ and the direction $\underline{\mathbf{N}}$.